

Safety Data Sheet



Product name: Vitamin AD3 1000/200 (Ethoxyquin free)

Product code: SLAD20

Current version: 1.2, issued: Feb.19,2019

Replaced version: 1.1, issued: Jan.03,2018

Section 1. Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier:

Product name:	Vitamin AD3 1000/200(Ethoxyquin free)
Product code:	SLAD20
Brand:	NHU
CAS No.:	127-47-9(Vitamin a acetate) 67-97-0(Vitamin D3)
EC No.:	204-844-2(Vitamin a acetate) 200-673-2(Vitamin D3)

1.2 Relevant identified uses of the substance or mixture and uses advised against:

Relevant identified uses of the substance or mixture:

Vitamins feed additives.

1.3 Details of the supplier of the safety data sheet:

Company

Zhejiang NHU CO. Ltd

Address

High-tech Industry Zone (Second),
Xinchang, Zhejiang, P. R. China, 312532

Tel: +86-575-86039932

Fax: +86-575-86165366

E-Mail: sales@cnhu.com

Website: <http://www.cnhu.com/>

1.4 Emergency telephone number:

Please contact the regional NHU representation in your country.

Section 2. Hazard identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 (CLP)

Repr.1B; H360

Aquatic Chronic 4; H413

For the full text of the H-Statements mentioned in this Section, see Section 16.

Classification according to Hazard Communication Standard (29 CFR 1910.1200)

Toxic to reproduction; Category 1B

Hazardous to the aquatic environment-Chronic; Category 4

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 (CLP Regulation)

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Hazard Pictograms



GHS08

Signal Word Danger

Hazardous component(s) to be indicated on label:

Vitamin A acetate

Hazard Statements

H360 May damage fertility or the unborn child.

H413 May cause long lasting harmful effects to aquatic life.

Precautionary statements

P201 Obtain special instructions before use.

P273 Avoid release to the environment.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P308+P313 IF exposed or concerned: Get medical advice/attention.

P501 Dispose of contents/container in accordance with local and national regulations.

Supplemental label elements

"Restricted to professional users"

Labelling according to Hazard Communication Standard (29 CFR 1910.1200):

Hazard Pictograms



Signal Word Danger

Hazard Statements

May damage fertility or the unborn child.

May cause long lasting harmful effects to aquatic life.

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Precautionary Statements

Obtain special instructions before use.

Do not handle until all safety precautions have been read and understood.

Avoid release to the environment.

Wear protective gloves/protective clothing/eye protection/face protection.

If exposed or concerned: Get medical advice/attention.

Store locked up.

Dispose of contents/container in corroding with local regulation.

2.3 Other hazards

Not available

Section 3. Composition and information on ingredients

3.1 Substances

Not applicable. The product is not a substance.

3.2 Mixtures

Chemical characterization

Vitamin A acetate and Vitamin D3 in gelatin.

Hazardous ingredients

No	Substance name			Additional information
	CAS/EC/Index No.	Classification(EC) 1272/2008(CLP)	Classification HCS 2012	Concentration (% ,b.w.)
1	Vitamin A acetate			
	127-47-9 204-844-2 -----	Repr.1B; H360D Aquatic Chronic.4; H413	Repr. 1B Aquatic Chronic.4; H413	36.0-41.0
2	Vitamin D3			
	67-97-0 200-673-2 603-180-00-4	Acute Tox, 2*;H330 Acute Tox, 3*;H311 Acute Tox, 3*;H301 STOT RE 1;H372**	Acute Tox. 2 (Inhalation) Acute Tox. 3 (Dermal) Acute Tox. 3(Oral) STOT R1	0.50-0.55
3	Cornstarch			
	9005-25-8 232-679-6 -----	----	----	20-30

Full text for all R-phrases, H-phrases and EUH-phrases: pls. see section 16

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(*, **, ***, ****) Detailed explanation pls. refer to CLP regulation No. 1272/2008, annex VI, 1.2

Additional ingredients

No	Substance name	CAS Number	Concentration(%,b.w.)
1	Glucose	5996-10-1	8-12
2	Gelatin	9000-70-8	25-35
3	Sodium ascorbate	134-03-2	1.0-3.0
4	Silicon dioxide	7631-86-9	0.2-1.0

Section 4. First aid measures

4.1 Description of first aid measures:

General advice:

In case of persisting adverse effects, consult a physician. Change contaminated, saturated clothing.

In case of inhalation:

Remove affected person from the immediate area. Ensure supply of fresh air. In case of respiratory arrest induce breathing with a respiratory device. Seek medical advice.

In case of eye contact:

Separate eyelids, wash the eyes thoroughly with water (15 min.). Seek medical assistance.

In case of skin contact:

Wash off immediately with plenty of soap and water. Consult a doctor if skin irritation persists.

In case of ingestion:

Seek medical advice immediately. Rinse mouth thoroughly with water. Never give anything by mouth to an unconscious person.

4.2 Most important symptoms and effects, both acute and delayed

No data available.

4.3 Indication of any immediate medical attention and special treatment needed

No data available.

Section 5. Fire-fighting measures

5.1 Extinguishing media:

Suitable extinguishing media :

Water spray jet, Foam, dry extinguishing powder or carbon dioxide.

Unsuitable extinguishing media:

No data available.

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5.2 Special hazards arising from the substance or mixture:

In the event of fire, the following can be released: Carbon dioxide (CO₂); Carbon monoxide (CO); Combustion products of this material have to be classed invariably as respiratory poison.

5.3 Advice for firefighters:

Use self-contained breathing apparatus. Wear protective clothing.

5.4 Further information:

No data available.

Section 6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures:

For non-emergency personnel

Refer to protective measures listed in sections 7 and 8. Ensure adequate ventilation. Avoid dust formation. Keep away sources of ignition.

For emergency responders

No data available. Personal protective equipment (PPE) - see Section 8.

6.2 Environmental precautions:

Do not discharge into the drains/surface waters/groundwater.

6.3 Methods and material for containment and cleaning up:

Take up mechanically. When picked up, treat material as prescribed under heading "Disposal considerations"

6.4 Reference to other sections:

No data available.

Section 7. Handling and storage

7.1 Precautions for safe handling:

Advice on safe handling

No special measures necessary if stored and handled as prescribed. Avoid the formation and deposition of dust. Provide good ventilation of working area (local exhaust ventilation, if necessary).

General protective and hygiene measures

Do not eat, drink or smoke during work time. Keep away from foodstuffs and beverages. Wash hands before breaks and after work. Provide eye wash fountain in work area. Do not inhale dust. It is essential for pregnant women to avoid inhaling the product and not to let it come in contact with the skin.

Advice on protection against fire and explosion

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Dust can form an explosive mixture with air. Take precautionary measures against static charges. Keep away from sources of heat and ignition.

7.2 Conditions for safe storage, including any incompatibilities:

Technical measures and storage conditions

Keep container tightly closed and dry in a cool, well-ventilated place.

Recommended storage temperature

Value <25°C

Requirements for storage rooms and vessels

Keep container hermetically closed in a dry, cool and well-ventilated place.

Always keep in containers of same material as the original one.

Advice on storage assembly

None known.

7.3 Specific end use(s):

No data available.

Section 8. Exposure controls/Personal protection

8.1 Control parameters:

INGREDIENT DATA:

Source	Ingredient	TWA	STEL	Peak	Notes
US OSHA Permissible Exposure Levels (PELS)-Table Z1	Cornstarch	15 mg/m3/ 5mg/m3	Not available	Not available	Total dust
US ACGIH Threshold Limit Values (TLV)	Cornstarch	10 mg/m3	Not available	Not available	TLV®Basis: Dermatitis
US NIOSH Recommended Exposure Limits(RELs)	Cornstarch	10 (total), 5 (resp)mg/m3	Not available	Not available	Not available
US OSHA Permissible Exposure Levels (PELs)- Table Z1	Silicon dioxide	Not available	Not available	Not available	See Table Z-3
US OSHA Permissible Exposure Levels	Silicon dioxide	Not available	Not available	Not available	See Table Z-3; containing less than 1%

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(PELs)- Table Z1					crystalline silica
US OSHA Permissible Exposure Levels (PELs)- Table Z3	Silicon dioxide	80/ (%SiO ₂) mg/m ³ / 20 mppcf	Not available	Not available	including natural diatomaceous earth
US NIOSH Recommended Exposure Limits (RELs)	Silicon dioxide	6 mg/m ³	Not available	Not available	Not available

EMERGENCY LIMITS:

Ingredient	TEEL-1	TEEL-2	TEEL-3
Glucose	77 mg/m ³	850 mg/m ³	5100mg/m ³
Gelatin	0.81 mg/m ³	8.9 mg/m ³	54 mg/m ³
Cornstarch	30 mg/m ³	1200 mg/m ³	1400mg/m ³
Silicon dioxide	6 mg/m ³	6 mg/m ³	6 mg/m ³
Silicon dioxide	6 mg/m ³	6 mg/m ³	630 mg/m ³
Silicon dioxide	18 mg/m ³	200 mg/m ³	1200 mg/m ³
Silicon dioxide	0.07 mg/m ³	0.77 mg/m ³	4.6 mg/m ³
Silicon dioxide	0.3 mg/m ³	0.3 mg/m ³	1.6 mg/m ³
Silicon dioxide	6 mg/m ³	6 mg/m ³	85 mg/m ³
Silicon dioxide	0.9 mg/m ³	9.9 mg/m ³	59 mg/m ³

Ingredient	Original IDLH	Revised IDLH
Vitamin A acetate	Not Available	Not Available
Vitamin D3	Not Available	Not Available
Glucose	Not Available	Not Available
Gelatin	Not Available	Not Available
Cornstarch	Not Available	Not Available
Sodium ascorbate	Not Available	Not Available
Silicon dioxide	N.E. mg/m ³ / N.E. ppm	3,000 mg/m ³

8.2 Exposure controls

Appropriate engineering controls:

No data available.

Individual protection measures, such as personal protective equipment:

Eye/face protection:

Safety glasses (EN 166).

Hand protection:

In case of intensive contact, wear protective gloves (EN 374). Sufficient protection is given wearing suitable protective gloves checked according to i.e. EN 374, in the event of risk of skin contact with the product. Before use, the protective gloves should be tested in any

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case for its specific work-station suitability (i.e. mechanical resistance, product compatibility and antistatic properties). Adhere to the manufacturer's instructions and information relating to the use, storage, care and replacement of protective gloves. Protective gloves shall be replaced immediately when physically damaged or worn. Design operations thus to avoid permanent use of protective gloves.

Respiratory protection:

If workplace exposure limits are exceeded, a respiration protection approved for this particular job must be worn.

Other

Normal chemical work clothing.

Environmental exposure controls:

No data available.

Section 9. Physical and chemical properties

9.1 Information on basic physical and chemical properties:

Physical state:

Solid

Form /Colour:

Microcapsule/ Brownish yellow to brownish.

Formula:

$C_{22}H_{32}O_2$ and $C_{27}H_{44}O$

Molecular Weight:

328.49 g/mol and 384.7 g/mol

Odour:

Slight odor

Odour threshold:

No data available

pH value:

No data available

Melting point/range(°C):

No data available

Boiling point/range(°C):

No data available

Decomposition point/decomposition range:

No data available

Flash point:

Value >93°C

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Auto-ignition temperature:

No data available

Flammability (solid, gas):

No data available

Lower flammability or explosive limits:

No data available

Upper flammability or explosive limits:

No data available

Vapour pressure:

No data available

Vapour density:

No data available

Evaporation rate:

No data available

Relative Density:

No data available

Solubility in water:

Practically insoluble.

Solubility (ies):

No data available

Partition coefficient: n-octanol /water:

No data available

Viscosity:

No data available

9.2 Other information:

Granularity :

100% Through Sieve No.20 (US)

Loss on drying:

≤5.0%

Percent Volatiles by Volume:

No data available

Bulk density:

No data available

Surface tension:

No data available

Explosive properties:

No data available

Oxidising properties:

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No data available

Section 10. Stability and reactivity

10.1 Reactivity:

No data available.

10.2 Chemical stability:

Stable under recommended storage conditions.

10.3 Possibility of hazardous reactions:

No data available.

10.4 Conditions to avoid:

Risk of dust explosions; Heat, naked flames and other ignition sources.

10.5 Incompatible materials:

Strong oxidizing agents; strong acids; strong bases.

10.6 Hazardous decomposition products:

Carbon monoxide and carbon dioxide

Section 11. Toxicological information

11.1 Information on toxicological effects

Acute oral toxicity (result of the ATE calculation for the mixture)	
No	Product Name
1	Vitamin AD3 1000/200 (Ethoxyquin free)
Remarks	The result of the applied calculation method according to the European Regulation (EC) 1272/2008 (CLP), Paragraph 3.1.3.6, Part 3 of Annex I is outside the values that imply a classification/labelling of this mixture according to table 3.1.1 defining the respective categories (ATE oral >2000 mg/kg)

Acute oral toxicity			
No	Substance Name	CAS no.	EC no.
1	Vitamin A acetate	127-47-9	204-844-2
LD50	> 2000 mg/kg bodyweight		
Species	rat		
with reference to	CAS 7069-42-3		
Method	OECD 401		
Source	ECHA		

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Acute dermal toxicity (result of the ATE calculation for the mixture)	
No	Product Name
1	Vitamin AD3 1000/200 (Ethoxyquin free)
Remarks	The result of the applied calculation method according to the European Regulation (EC) 1272/2008 (CLP), Paragraph 3.1.3.6, Part 3 of Annex I is outside the values that imply a classification/labelling of this mixture according to table 3.1.1 defining the respective categories (ATE dermal > 2000 mg/kg)

Acute dermal toxicity
No data available

Acute inhalational toxicity (result of the ATE calculation for the mixture)	
No	Product Name
1	Vitamin AD3 1000/200 (Ethoxyquin free)
Remarks	The result of the applied calculation method according to the European Regulation (EC) 1272/2008 (CLP), Paragraph 3.1.3.6, Part 3 of Annex I is outside the values that imply a classification/labelling of this mixture according to table 3.1.1 defining the respective categories (ATE for inhalation: > 20.000 ppmV (gases), > 20 mg/l (vapours), > 5mg/l (dusts/mists)).

Acute inhalational toxicity
No data available
Skin corrosion/irritation
No data available
Serious eye damage/irritation
No data available
Respiratory or skin sensitisation
No data available
Germ cell mutagenicity
No data available
Reproduction toxicity
No data available
Carcinogenicity
No data available
STOT-single exposure
No data available

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STOT-repeated exposure
No data available
Aspiration hazard
No data available
Symptoms related to the physical, chemical and toxicological characteristics
Eye contact may cause mechanical irritation through dust particles.
Delayed and immediate effects as well as chronic effects from short and long-term exposure
Inhalation of dusts may irritate the respiratory tract.

Section 12. Ecological information

12.1 Toxicity:

Toxicity to fish (acute)
No data available
Toxicity to fish (chronic)
No data available
Toxicity to Daphnia (acute)
No data available
Toxicity to Daphnia (chronic)
No data available
Toxicity to algae (acute)
No data available
Toxicity to algae (chronic)
No data available
Bacteria toxicity
No data available

12.2 Persistence and degradability:

Biodegradability			
No	Substance Name	CAS no.	EC no.
1	Vitamin A acetate	127-47-9	204-844-2
Duration		28 day(s)	
Method		OECD 301 B	
Source		ECHA	
Evaluation		Not readily biodegradable	

12.3 Bioaccumulative potential:

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No data available

12.4 Mobility in soil:

No data available

12.5 Results of PBT & vPvB assessment:

No data available

12.6 Other adverse effects:

No data available

12.7 Other information:

Do not let enter the product into drains or waterways and do not store on public depositories.

Section 13. Disposal considerations

13.1 Waste treatment methods:

Product:

Allocation of a waste code number, according to the European Waste Catalogue, should be carried out in agreement with the regional waste disposal company.

Packaging:

Residuals must be removed from packaging and when emptied completely disposed of in accordance with the regulations for waste removal. Incompletely emptied packaging must be disposed of in the form of disposal specified by the regional disposer.

Section 14. Transport information

14.1 Transport ADR/RID/ADN

The product is not subject to ADR/RID/AND regulations.

14.2 Transport IMDG

The product is not subject to IMDG regulations.

14.3 Transport ICAO-TI/IATA

The product is not subject to ICAO-TI/IATA regulations.

14.4 Other information

No data available

14.5 Environmental hazards

Information on environmental hazards, if relevant, please see 14.1-14.3.

14.6 Special precautions for user

No data available

14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not relevant

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Section 15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU regulations

Regulation (EC) No 1907/2006 (REACH) Annex XIV (List of substances subject to authorisation)

According to the data available and/or specifications supplied by upstream suppliers, this product does not contain any substances considered as substances requiring authorisation as listed on Annex XIV of the REACH regulation (EC) 1907/2006.

REACH candidate list of substances of very high concern (SVHC) for authorisation

According to available data and the information provided by preliminary suppliers, the product does not contain substances that are considered substances meeting the criteria for inclusion in annex XIV (List of Substances Subject to Authorisation) as laid down in Article 57 and article 59 of REACH (EC) 1907/2006.

Regulation (EC) No 1907/2006 (REACH) Annex XVII: RESTRICTIONS ON THE MANUFACTURE, PLACING ON THE MARKET AND USE OF CERTAIN DANGEROUS SUBSTANCES, PREPARATIONS AND ARTICLES

According to the data available and/or specifications supplied by upstream suppliers, this product does not contain any substances requiring authorisation as listed on Annex XIV of the REACH regulation (EC) 1907/2006.

Restriction of occupation

Observe employment restrictions for child bearing mothers and nursing mothers.

Council Directive 96/82/EC on the control of major-accident hazards involving dangerous substances

Remarks Annex I, part 1+2: not mentioned. With regard to possibly appropriate decomposition products see Chapter 10.

US regulations

Vitamin A acetate(127-47-9) is found on the following regulatory lists	"US Toxic Substances Control Act (TSCA) - Chemical Substance Inventory" List.
Vitamin D3 (67-97-0) is found on the following regulatory lists	"US Toxic Substances Control Act (TSCA) - Chemical Substance Inventory" List.
Glucose(5996-10-1) is found on the following regulatory lists	"US Toxic Substances Control Act (TSCA) - Chemical Substance Inventory" List.
Gelatin (9000-70-8) is found on the following regulatory lists	"US Toxic Substances Control Act (TSCA) - Chemical Substance Inventory" List.

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Cornstarch(9005-25-8) is found on the following regulatory lists	"US - Hawaii Air Contaminant Limits" List. "US - Idaho - Limits for Air Contaminants" List. "US -Alaska Limits for Air Contaminants" List. "US -Wyoming Toxic and Hazardous Substances Table Z1 Limits for Air Contaminants" List. "US Toxic Substances Control Act (TSCA) -Chemical Substance Inventory" List.
Silicon dioxide (7631-86-9) is found on the following regulatory lists	"US - Idaho - Toxic and Hazardous Substances- Mineral Dust" List. "US - Hawaii Air Contaminant Limits" List. "US - Wyoming Toxic and Hazardous Substances Table Z-3 Mineral Dusts" List. "US - Idaho - Limits for Air Contaminants" List. "US - Washington Toxic air pollutants and their ASIL, SQER and de minimis emission values" List. "US - Alaska Limits for Air Contaminants" List. "US Toxic Substances Control Act (TSCA) – Chemical Substance Inventory" List.

15.2 Chemical safety assessment

No data available.

Section 16. Other information

16.1 Full text of R-, H- and EUH-phrases drawn up in sections 2 and 3 (provided not already drawn up in these sections):

Repr.	Reproductive toxicity
Acute Tox.	Acute toxicity
H301	Toxic if swallowed.
H311	Toxic in contact with skin.
H330	Fatal if inhaled.
H372	Causes damage to organs through prolonged or repeated exposure.
H360D	May damage the unborn child.
H413	May cause long lasting harmful effects to aquatic life.
STOT RE	Specific target organ toxicity – repeated exposure.

16.2 Other information:

HMIS® ratings

Health: 4
Flammability: 1
Physical hazard: 0

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NFPA ratings

Health: 4
Flammability: 1
Instability: 0

16.3 Further information

The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. This safety data sheet should be used in conjunction with technical sheets. It does not replace them. The information is based on our current knowledge however it does not represent a guarantee of product properties nor does it create any legal obligation. NHU Corporation shall not be held liable for any damage resulting from handling or from contact with the above product.

ORIGINAL CONTROLLED

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QA Manager
胡柏藩